

Do LLMs Forget Historical Context?

Historical-Sense Identification with Retrieved Lexical Evidence

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Abstract

This project studies whether language models can identify historically appropriate word senses when those senses differ from present-day English. I formulate a controlled historical-sense identification task: given a passage, a target expression, and a historical period, the model must identify the period-appropriate sense of the target expression. The initial pilot showed that generic historical prompting does not reliably reduce modern-sense leakage. I then extended the setup toward a retrieval-augmented method: a retrieval module attaches historical lexical evidence to each example, and the model answers with that evidence in context. On the current 12-example Early Modern English dataset, baseline prompting produced 1 correct answer, generic historical prompting produced 0 correct answers, retrieved lexical evidence produced 4 correct and 3 partial answers, and an oracle manually curated glossary condition produced 9 correct and 3 partial answers. The retrieval prototype improves over generic prompting but remains far below the oracle condition, revealing an evidence-use gap: even when relevant lexical evidence is retrieved, the model may ignore it, use distractors, or drift back toward modern explanation.

1 Introduction

Large language models can produce fluent explanations of historical texts while relying on modern meanings of individual words. This project studies that failure mode as *modern-sense leakage*. For example, in Early Modern English, *anon* can mean “right away,” *quick* can mean “living,” and *conversation* can mean “conduct” or “way of life.” A model that interprets these expressions as “anonymous,” “fast,” or spoken dialogue is applying a modern sense where a historical one is required.

The project began as a pilot on period-constrained interpretation. I have now narrowed it into a more concrete task and started extending it into a method-oriented study. The current question is not only whether models make historical sense errors, but whether lexical evidence retrieval can reduce those errors.

This draft makes three contributions. First, it defines a small but controlled historical-sense identification task that isolates word-sense errors from broader literary interpretation. Second, it compares four prompting conditions: no historical support, generic historical instruction, retrieved lexical evidence, and an oracle glossary note. Third, it separates retrieval success from evidence-use success. In the current run, retrieval places the correct headword first for every example, but the model still often fails to use that evidence. This suggests that historical RAG systems should be evaluated not only by whether they retrieve relevant evidence, but also by whether the model’s final answer actually follows that evidence.

2 Related Work and Positioning

This project is related to computational work on lexical semantic change, retrieval-augmented generation, and the evaluation of language-model meaning. Lexical semantic change detection studies how word meanings differ across time periods. SemEval-2020 Task 1 framed this as an unsupervised task over diachronic corpora and emphasized the need for manually annotated evaluation data [4]. My task is narrower: instead of detecting whether a word changed meaning in general, it asks whether a model can identify the period-appropriate sense of a target expression in a specific passage.

The retrieval component follows the broad motivation of retrieval-augmented generation: external evidence can supply information that a parametric model may not access reliably [3]. However, most RAG evaluations focus on factual or knowledge-intensive question answering. Here the retrieved material is lexical and historical. The central question is not only whether the evidence is present, but whether the model resists a familiar modern sense when the retrieved evidence points elsewhere.

The project also connects to critiques of treating fluent language-model output as evidence of robust understanding. Bender and Koller argue that models trained only on form should not automatically be assumed to capture meaning [1]. Modern-sense leakage is a concrete version of this problem: the answer can sound plausible while assigning the wrong historically situated sense. Finally, the next retrieval source is motivated by Lexicons of Early Modern English (LEME), which provides historical lexical works from roughly 1475–1755 and makes many word entries and encoded texts available for research use [2].

3 Task

The task is historical-sense identification. Each example contains a historical period, source, passage, target expression, expected historical interpretation, common modern failure interpretation, and rationale. The model is asked to identify the historical sense of the target expression in context.

The current dataset has 12 hand-curated Early Modern English examples from Shakespeare, general Early Modern usage targets, and the King James Bible.

ID	Target	Historical sense
rj_wherefore	wherefore	why
hamlet_soft	soft	wait; hold a moment
rj_anon	anon	right away; in a moment
eme_presently	presently	immediately; very soon
eme_brave	brave	fine; splendid; showy
eme_conceit	conceit	thought; idea; mental conception
tempest_admiration	admiration	wonder; astonishment
kjv_suffer	suffer	allow; permit
kjv_prevent	prevent	precede; go before
kjv_conversation	conversation	conduct; way of life
kjv_quick	quick	living; alive
kjv_charity	charity	Christian love

Table 1: Current historical-sense identification dataset.

4 Compressed Pilot Findings

The first stage of the project built a Modal pipeline and evaluated `Qwen/Qwen2.5-0.5B-Instruct` on a small pilot. The initial 8-example experiment compared baseline prompting against a generic historical prompt. The baseline produced 1 correct, 3 partial, and 4 incorrect answers; the historical prompt produced 1 correct, 1 partial, and 6 incorrect answers. This showed that a lightweight instruction to think historically was not enough.

I then expanded the dataset to 12 examples and tried an LLM-as-judge evaluation pass. That judge was not reliable: it labeled all outputs as partial and all modern-leakage labels as no leakage, even when the model interpreted *anon* as anonymous, *quick* as fast, and *conversation* as dialogue. Because the automatic judge failed to distinguish the target errors, the current results use manual labels for historical accuracy and modern leakage.

The next completed pilot comparison added an oracle glossary condition, where the prompt included the manually curated historical sense. This condition was much stronger: baseline prompting gave 1 correct answer, generic historical prompting gave 0 correct answers, and the oracle glossary condition gave 9 correct and 3 partial answers. That result suggested that targeted lexical evidence matters more than generic historical instruction.

5 Retrieval-Augmented Method Prototype

The new method experiment asks whether the oracle glossary improvement can be approached by an automatic retrieval pipeline. The implemented prototype separates the task into two stages:

1. retrieve historical lexical evidence for the target expression and period;
2. condition the model on the retrieved evidence when answering the historical-sense question.

For this prototype, I built a small lexical evidence index with one seed entry for each current target word, plus top- k retrieval that returns the three highest-scoring entries. Each evidence entry contains a headword, period, source field, date field, and gloss. The retrieved dataset stores these entries under a `retrieved_evidence` field, so the evaluation script can use either retrieved evidence or oracle evidence without changing the original examples.

Retrieval is intentionally simple in the current draft. The target expression and entry headword are normalized by lowercasing and retaining alphanumeric tokens. A candidate receives a large score bonus when its normalized headword exactly matches the target, a smaller bonus for partial headword overlap, a period-match bonus, and a small token-overlap score between the example text and the evidence entry. This simple retriever is not meant to be the final method. Its purpose is to create an interpretable first test of whether top- k lexical evidence helps and where failures occur.

The retrieved prompt has the following form:

Retrieved historical lexical evidence:

– headword (period, source): gloss

Use the retrieved historical lexical evidence when it is relevant, but do not assume every retrieved note is correct.

This is different from the oracle glossary prompt. The oracle prompt gives only the manually curated historical interpretation for the target expression. The retrieved prompt gives top- k evidence, which may include distractors. This tests whether the model can use retrieved lexical evidence selectively rather than simply copy a gold gloss.

6 Experimental Setup

All current experiments use `Qwen/Qwen2.5-0.5B-Instruct` and the same 12-example dataset. The four evaluated conditions are:

- **Baseline:** no historical support.
- **Historical prompt:** generic instruction to prefer Early Modern English senses.
- **Retrieved evidence:** top- k lexical evidence retrieved automatically from the seed evidence index.
- **Oracle glossary:** manually curated historical sense from the dataset.

Evaluation uses manual labels:

- **Historical accuracy:** correct, partial, or incorrect.
- **Modern leakage:** no, mixed, or yes.

I use manual evaluation because the initial LLM-as-judge run collapsed all outputs into the same label and missed obvious historical-sense errors. A response is labeled *correct* when it gives the expected historical sense in context, *partial* when it gestures toward the historical sense but remains vague or mixes in a modern reading, and *incorrect* when it misses the historical sense. Modern leakage is labeled *yes* when the answer clearly defaults to the modern meaning, *mixed* when both historical and modern interpretations appear, and *no* when the answer avoids the modern failure mode.

7 Results

Condition	Correct	Partial	Incorrect
Baseline	1	4	7
Historical prompt	0	2	10
Retrieved evidence	4	3	5
Oracle glossary	9	3	0

Table 2: Manual historical-accuracy results.

Condition	No leakage	Mixed leakage	Leakage yes
Baseline	5	2	5
Historical prompt	4	1	7
Retrieved evidence	9	1	2
Oracle glossary	10	2	0

Table 3: Manual modern-leakage results.

The retrieved-evidence condition improves over the generic historical prompt: correct answers increase from 0 to 4, and clear modern leakage decreases from 7 cases to 2. However, retrieved

evidence is still far below the oracle glossary condition, which produces 9 correct answers and no fully incorrect answers.

The retrieval module itself found the correct headword as the top-ranked evidence item for all 12 examples in this prototype. Therefore, the main failures in this run are not retrieval misses. They are evidence-use failures: the correct evidence is present, but the model sometimes ignores it, uses a distractor, or gives an unstable explanation.

Table 4 makes this gap explicit. In the retrieved condition, lexical retrieval succeeds on every example by the top-1 headword-hit metric, but answer accuracy remains far lower than the oracle glossary condition. The problem is therefore downstream of lexical lookup: the model receives useful evidence but does not reliably bind that evidence to the target expression in the passage.

Metric	Retrieved evidence	Oracle glossary
Top-1 correct-headword retrieval	12/12	N/A
Correct historical answers	4/12	9/12
Correct or partial answers	7/12	12/12
Clear modern leakage	2/12	0/12

Table 4: Retrieval success versus final answer success.

8 Error Analysis

Anon. The correct retrieved evidence for *anon* says “right away; in a moment; soon.” However, because the prompt includes top- k retrieved evidence, it also includes distractors. The model answered that the historical sense of *anon* was “brave,” apparently copying an irrelevant retrieved entry. This is not a retrieval-hit failure; it is a failure to select the right evidence from the retrieved context.

Conversation. For *conversation*, the expected historical sense is conduct or way of life. The retrieved evidence included the correct entry, but the model answered with an irrelevant “right away” explanation. This is another distractor-use failure.

Soft. For “Soft you now,” the retrieved evidence included “wait; hold a moment; stop.” The model nevertheless answered with a modern meaning related to gentleness or soothing tone. This is a direct modern-sense leakage error despite the presence of correct evidence.

Brave, suffer, quick, and charity. The retrieval condition worked well on these examples. It identified *brave* as fine or showy, *suffer* as allow or permit, *quick* as living or alive, and *charity* as Christian or selfless love.

These examples suggest three failure categories for the next version of the project:

- **Modern-default failures:** the model ignores relevant evidence and returns the present-day sense, as with *soft* and *presently*.
- **Distractor-selection failures:** the model copies an irrelevant retrieved entry, as with *anon* and *conversation*.

- **Vague-evidence failures:** the model moves in the right direction but does not commit to the historically specific sense, as with *conceit* and *prevent*.

This taxonomy matters because the fixes are different. Modern-default failures may require stronger instruction or contrastive examples. Distractor-selection failures point to prompt formatting, re-ranking, or forcing the answer to cite the evidence item used. Vague-evidence failures suggest the need for a stricter answer format or a multiple-choice sense-selection setup before free-form explanation.

9 Discussion

The current result changes the project from a toy prompt comparison into a method question. The oracle glossary result shows that lexical evidence can strongly improve historical-sense identification. The retrieval prototype shows that automatically supplied evidence can help, but also reveals a new problem: evidence use is not guaranteed.

This distinction is important. A RAG system can fail because it retrieves the wrong evidence, but it can also fail because the model receives the right evidence and still uses it badly. In this prototype, top-1 retrieval hits the correct headword for every example, yet the model only reaches 4 correct answers. The gap between retrieved evidence and oracle glossary is therefore not only a retrieval-quality issue; it is also a context-use and distractor-resistance issue.

The current evidence also weakens a simpler “historical prompting is enough” interpretation. The generic historical prompt performs worse than the baseline in historical accuracy on this dataset, producing 0 correct answers and 10 incorrect answers. This does not prove that all historical prompting is ineffective, but it does show that a broad instruction to prefer Early Modern senses is not a reliable substitute for target-specific lexical evidence.

10 Limitations and Next Experimental Step

This is still a small prototype. The evidence index is a seed index rather than a full historical dictionary or corpus. The dataset has only 12 examples, and only one model has been evaluated so far. These limits mean the current numbers should not be presented as general benchmarks.

The next experiment should replace the seed evidence index with entries from historical lexical resources such as Lexicons of Early Modern English, Shakespeare glossaries, and KJV historical glossaries. The dataset should also be expanded to roughly 50–100 examples, with each example linked to a retrievable evidence source. After that, the same four-condition comparison should be run across multiple models.

The immediate method changes should be:

1. add source-linked lexical entries rather than hand-written seed entries;
2. report retrieval hit rate separately from answer accuracy;
3. add an evidence-use label indicating whether the final answer used the correct retrieved entry;
4. test a stricter retrieved prompt that asks the model to name which evidence item supports its answer;
5. run at least one stronger model to check whether the evidence-use gap is mainly a small-model limitation.

11 Conclusion

The project now has two completed findings. First, generic historical prompting does not reliably reduce modern-sense leakage in the current historical-sense identification task. Second, lexical evidence helps, but the way evidence is provided matters. Oracle glossary evidence is highly effective, while automatically retrieved top- k evidence produces a smaller improvement and introduces distractor-use failures.

The next methodological focus is therefore clear: build a stronger historical lexical retrieval pipeline and evaluate not only whether it retrieves the correct evidence, but whether the model can actually use that evidence to avoid modern-sense leakage.

References

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